



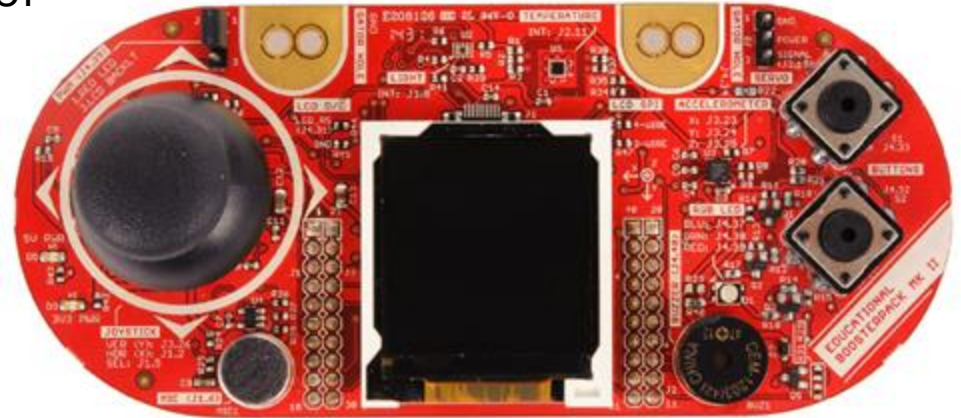
Embedded Software for the Internet Of Things

MSP432 – BOOSTERPACK SENSORS

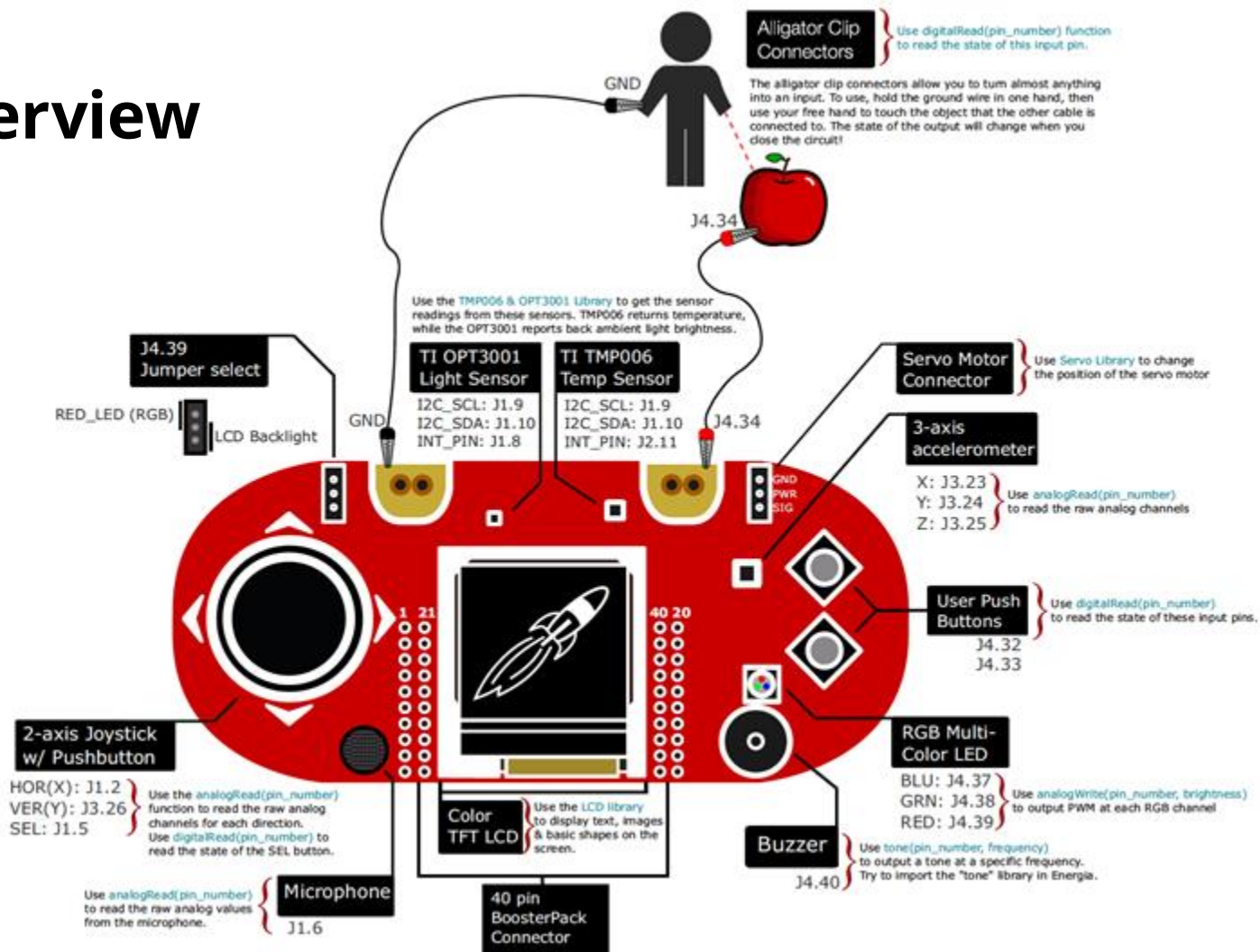
Assoc. Prof. Kasim Sinan Yildirim
University of Trento, Italy

BOOSTXL-EDUMKII EDUCATIONAL BOOSTERPACK

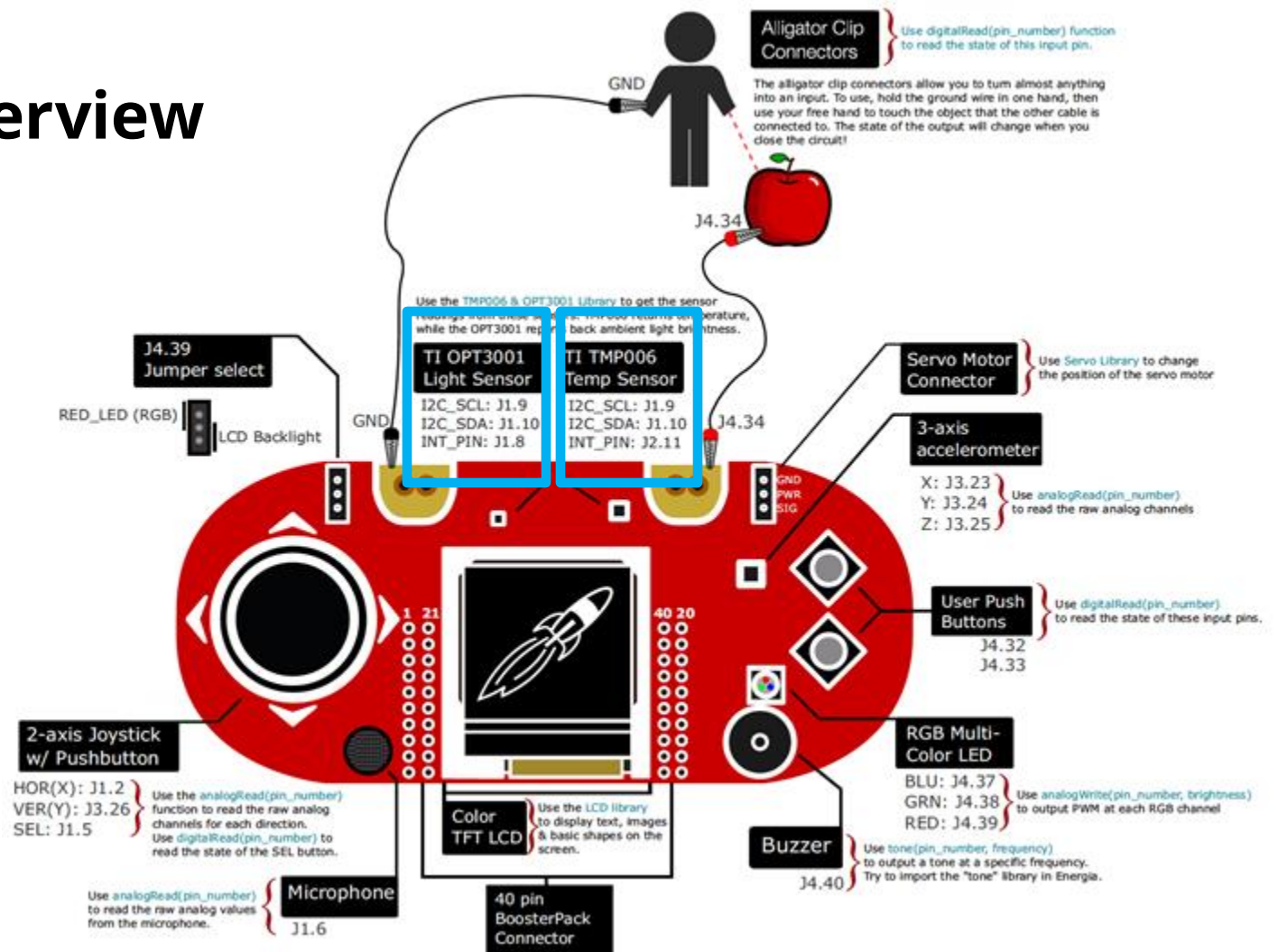
- Various analog and digital inputs and outputs
- TI OPT3001 Light Sensor
- TI TMP006 Temperature Sensor
- Servo Motor Connector
- 3-Axis Accelerometer
- User Push Buttons
- RGB Multi-color LED
- Buzzer
- Color TFT LCD Display
- Microphone
- 2-Axis Joystick with Pushbutton



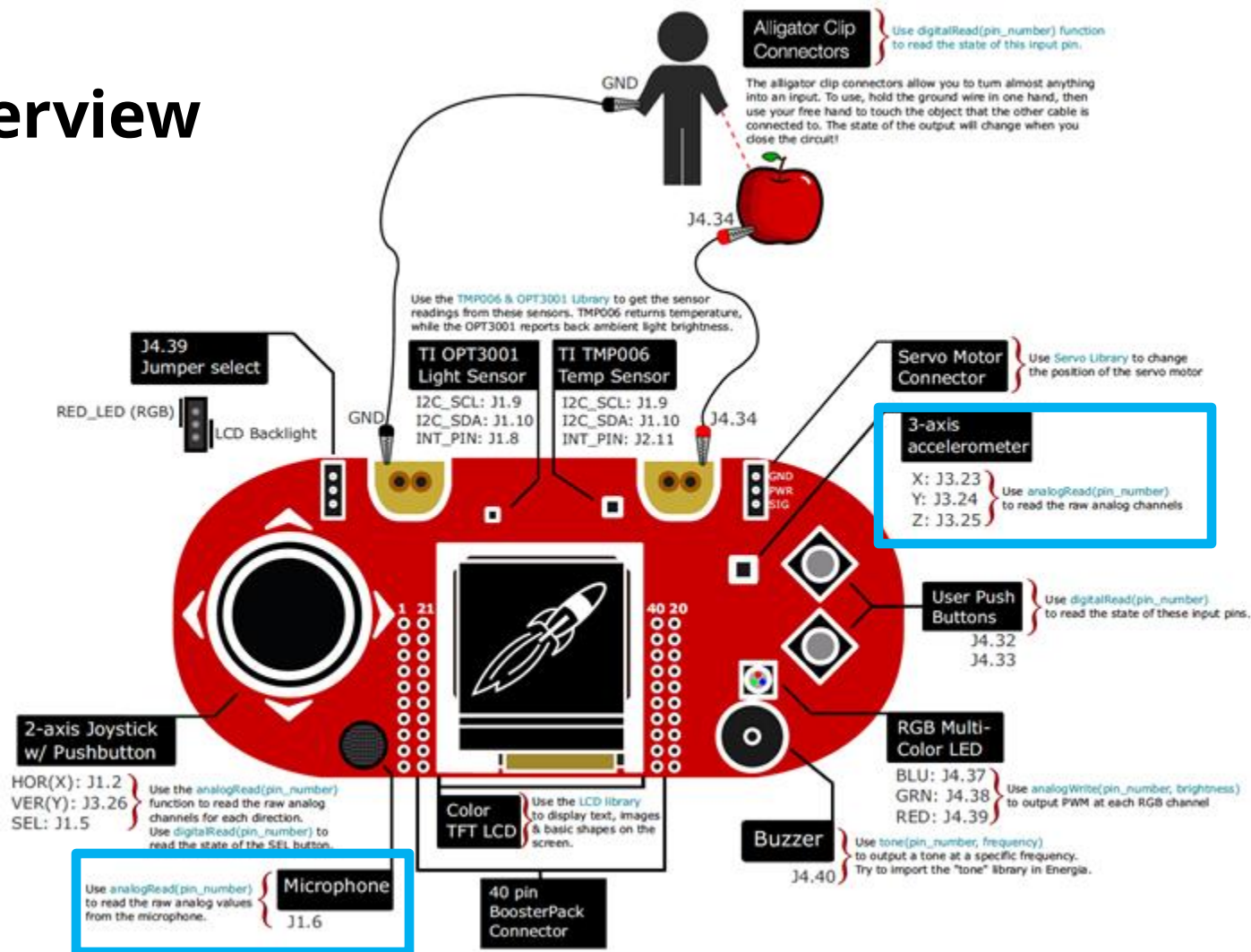
Sensor Overview



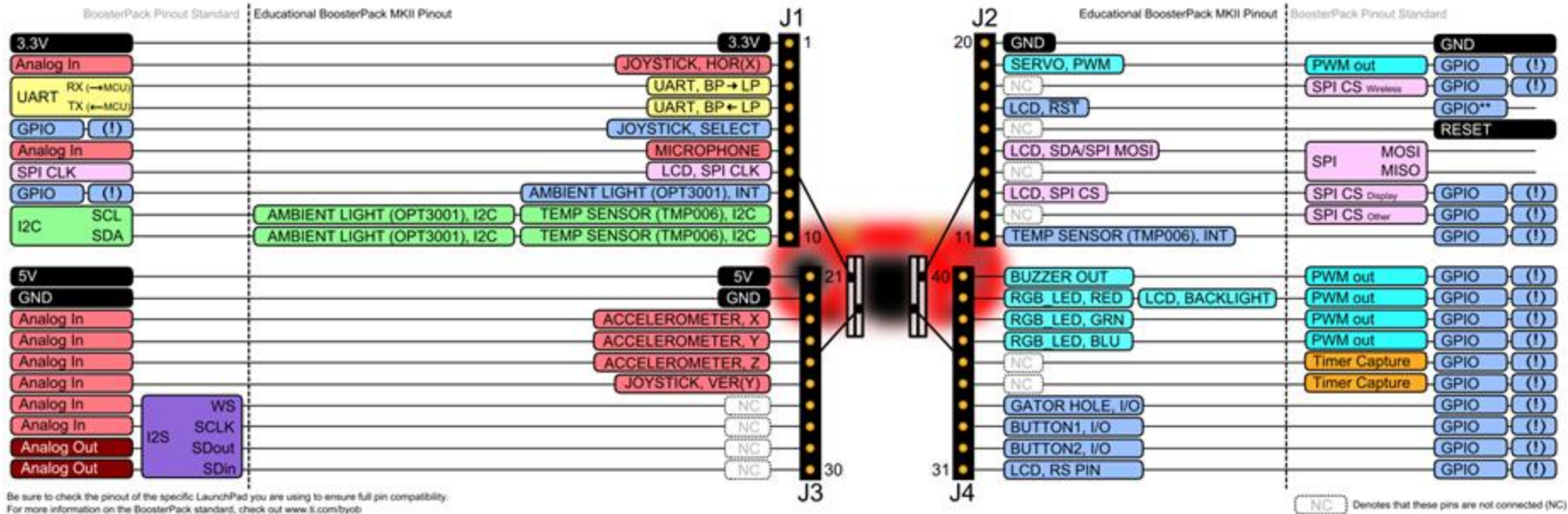
Sensor Overview



Sensor Overview



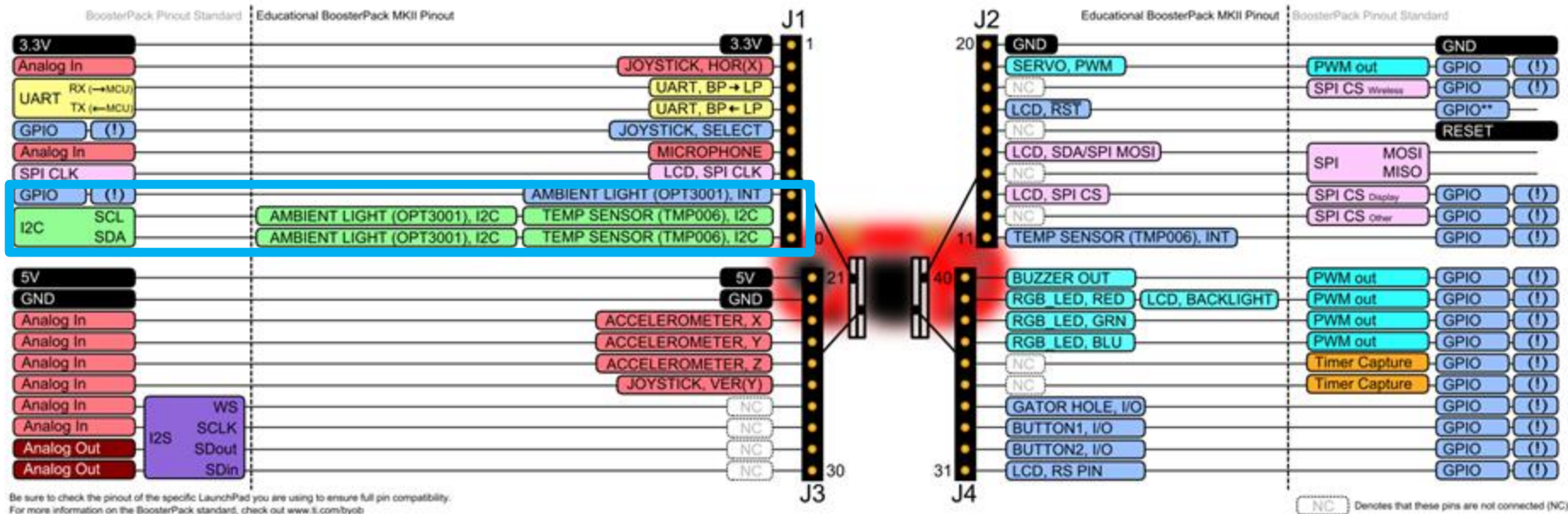
PinOut Diagram



Be sure to check the pinout of the specific LaunchPad you are using to ensure full pin compatibility. For more information on the BoosterPack standard, check out www.ti.com/tyob

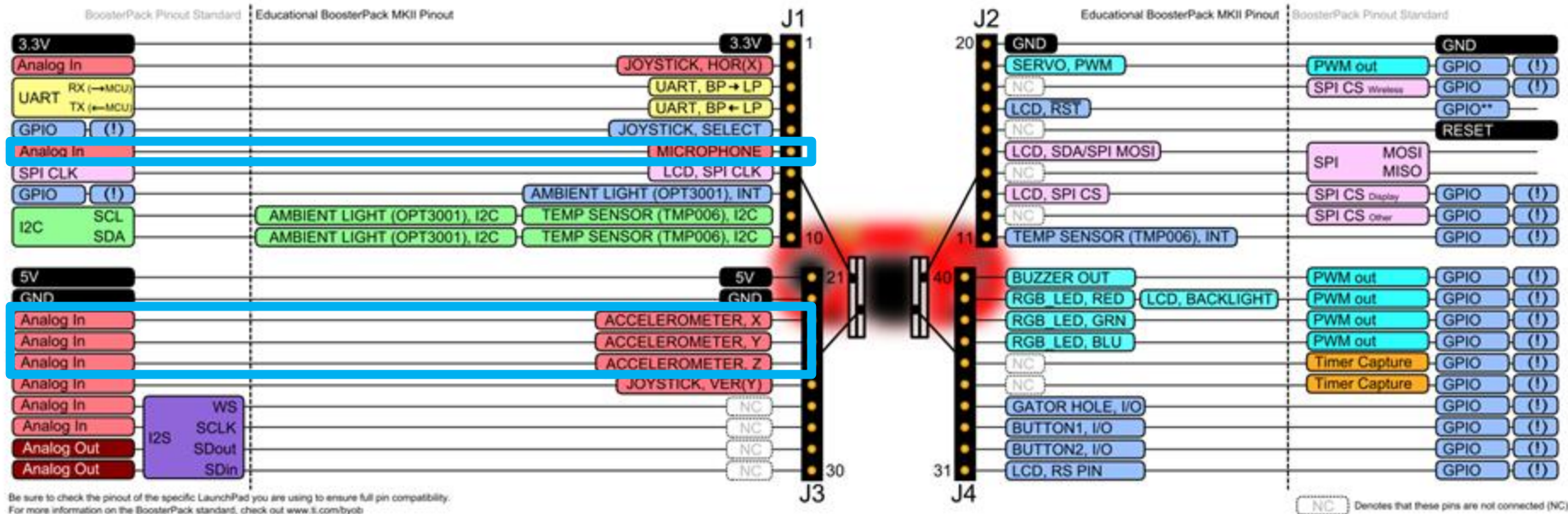
NC Denotes that these pins are not connected (NC)

PinOut Diagram



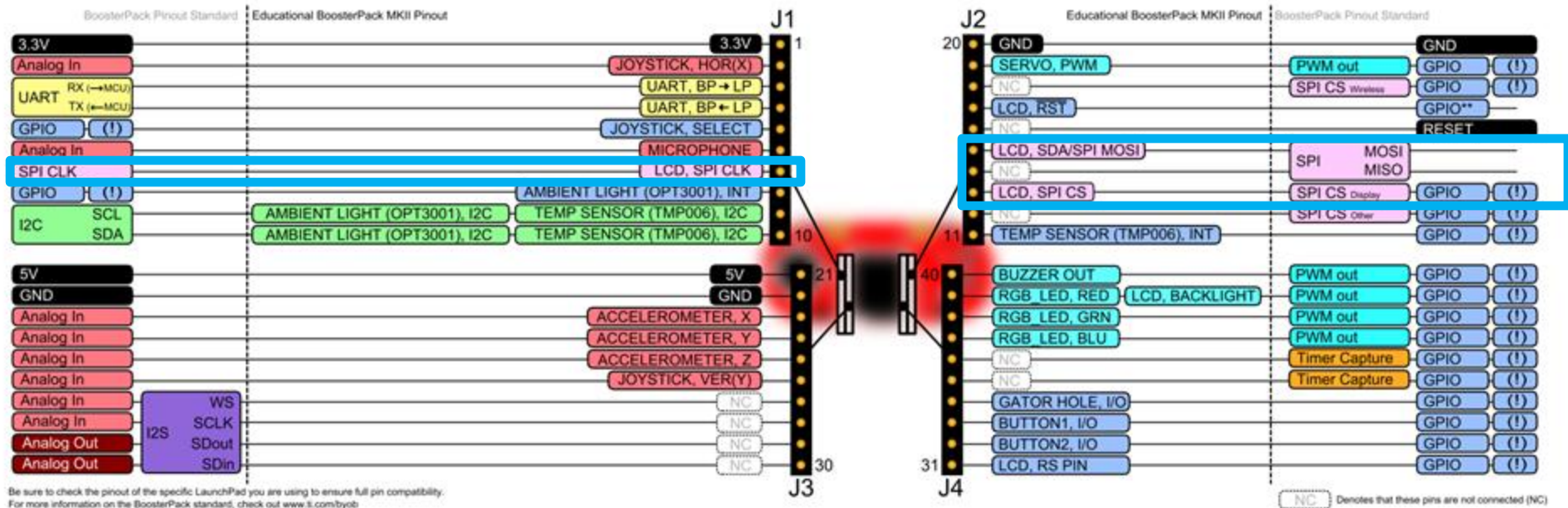
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PinOut Diagram



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PinOut Diagram

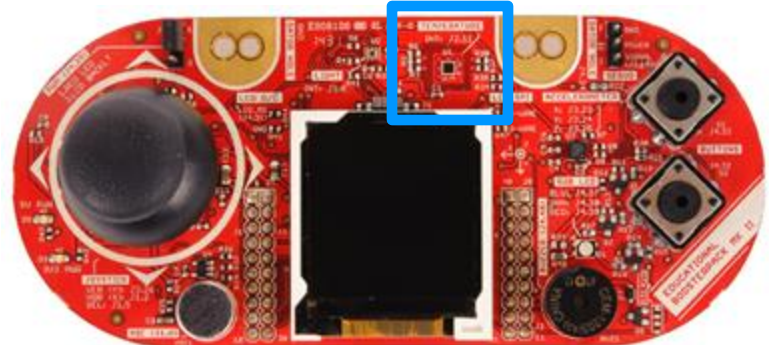


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TMP006 INFRARED CONTACTLESS TEMPERATURE SENSOR

- Measures the temperature without the need to make contact with the object.
 - uses a thermopile to absorb the infrared energy emitted from the object
 - *A thermopile is an electronic device that converts thermal energy into electrical energy.*
 - *the change in thermopile voltage to determines the object temperature.*

BoosterPack Plug-in Module Header Connection	Pin Function
J1.9 ⁽¹⁾	I ² C SCL
J1.10 ⁽¹⁾	I ² C SDA
J2.11	TMP006 Interrupt

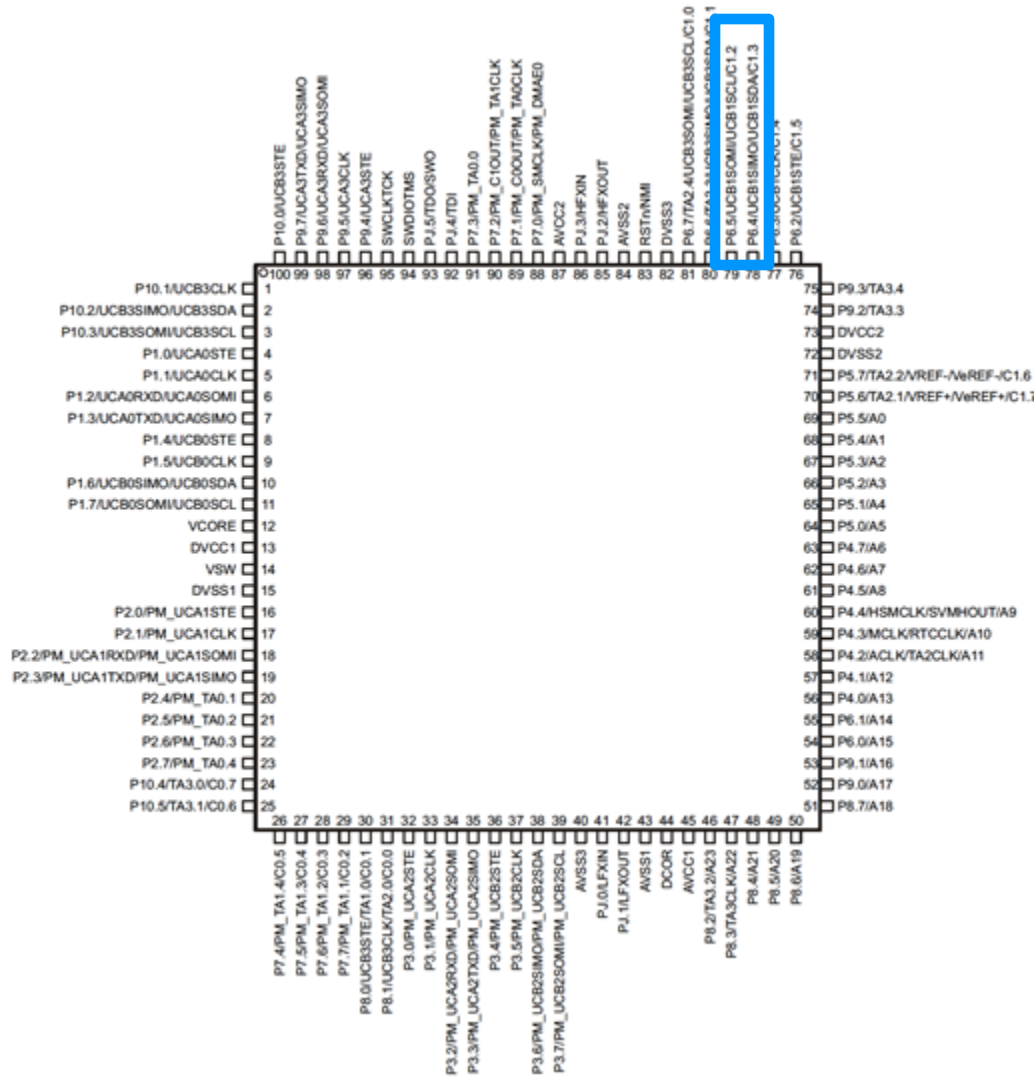


Pin Mappings

- Launchpad pins:
 - P6.5: I2C_SCL
 - P6.4: I2C_SDA



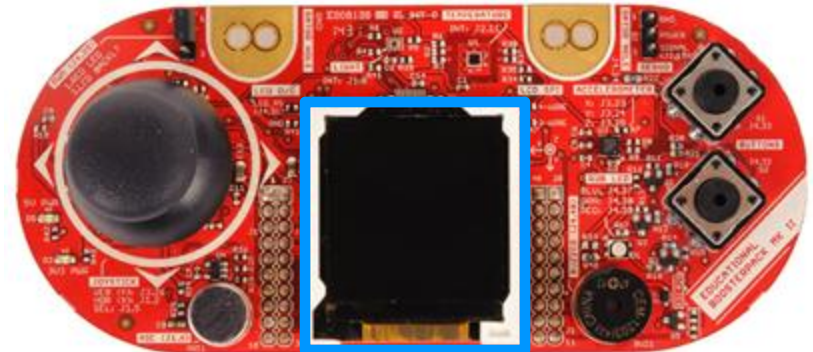
PIN NAME (P6.x)	x	FUNCTION	CONTROL BITS OR SIGNALS ⁽¹⁾		
			P6DIR.x	P6SEL1.x	P6SEL0.x
P6.2UCB1STE/C1.5 ⁽²⁾	2	P6.2 (I/O)	I: 0; O: 1	0	0
		UCB1STE	X ⁽³⁾	0	1
		N/A	0	1	0
		DVSS	1	1	0
P6.3UCB1CLK/C1.4 ⁽²⁾	3	C1.5 ⁽⁴⁾⁽⁵⁾	X	1	1
		P6.3 (I/O)	I: 0; O: 1	0	0
		UCB1CLK	X ⁽³⁾	0	1
		N/A	0	1	0
		DVSS	1	1	0
P6.4UCB1SIMO/UCB1SDA/C1.3 ⁽²⁾	4	C1.4 ⁽⁴⁾⁽⁵⁾	X	1	1
		P6.4 (I/O)	I: 0; O: 1	0	0
		UCB1SIMO/UCB1SDA	X ⁽³⁾	0	1
		N/A	0	1	0
P6.5UCB1SOMI/UCB1SCL/C1.2 ⁽²⁾	5	C1.3 ⁽⁴⁾⁽⁵⁾	X	1	1
		P6.5 (I/O)	I: 0; O: 1	0	0
		UCB1SOMI/UCB1SCL	X ⁽³⁾	0	1
		N/A	0	1	0
		DVSS	1	1	0
		C1.2 ⁽⁴⁾⁽⁵⁾	X	1	1



LCD Display

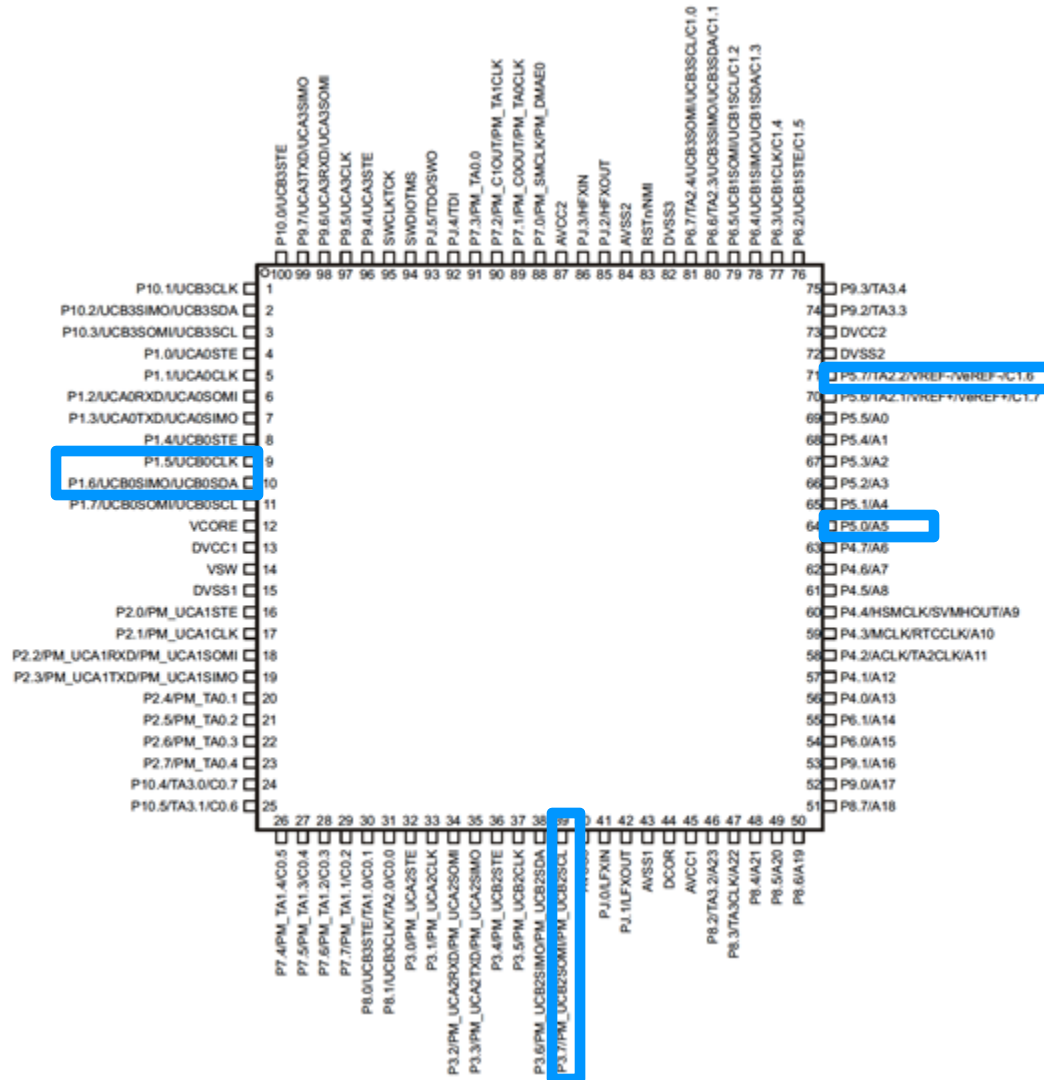
- 128x128-pixel TFT LCD
 - supports display updates up to 20 frames per second (FPS)
 - controlled through the SPI interface
 - has a color depth of 262K colors and a contrast ratio of 350

BoosterPack Plug-in Module Header Connection	Pin Function
J1.7	LCD SPI clock
J2.13	LCD SPI chip select
J2.15	LCD SPI MOSI
J4.31	LCD reset pin
J4.39 ⁽¹⁾	LCD backlight



Pin Mapping

- SCK_PORT: P1.5
- MOSI_PORT: P1.6
- RST_PORT: P5.7
- CS_PORT: P5.0
- DC_PORT: P3.7



Lab #1

- Perform temperature sensing via timers
 - Set a 1 second timer and update the LCD accordingly.
 - Display the temperature in Celsius.